C27 Aromatic Chemistry – Knowledge Organiser

AQA A-level Chemistry • Set 1: Benzene **Paper 2**

#	Question	Answer
1	Molecular formula of benzene	C_6H_6
2	Bond angle in benzene	120°
3	C–C bond length in benzene	Between C–C and C=C
4	Bonding in benzene	Ring of delocalised π electrons
5	Enthalpy of hydrogenation of benzene vs Kekulé	Less exothermic than expected
6	Expected enthalpy of hydrogenation of cyclohexa-1,3,5-triene	$-360\ kJ\ mol^{-1}$
7	Why benzene undergoes substitution not addition	Keeps stable delocalised ring
8	Mechanism type for benzene	Electrophilic substitution
9	Name of the C_6H_5- group	Phenyl
10	Why benzene is resistant to addition	Delocalisation energy (stability)

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AQA A-level Chemistry • Set 2: Nitration and Acylation **Paper 2**

#	Question	Answer
1	Reagents for nitration	Conc HNO_3 + conc H_2SO_4
2	Temperature for mononitration	Below 55 °C
3	Electrophile in nitration	NO_2^+
4	Equation forming NO_2^+	$HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2HSO_4^-$
5	Uses of nitration	Explosives; making amines for dyes
6	Reagents for Friedel–Crafts acylation	Acyl chloride + $AlCl_3$
7	Electrophile in acylation	RCO^+
8	Equation forming the acylium ion	$RCOCl + AlCl_3 \rightarrow RCO^+ + AlCl_4^-$
9	Equation regenerating the catalyst	$H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$
10	Product of benzene + ethanoyl chloride	Phenylethanone